Zihao Wang

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EDUCATION

Nanyang Technological University, Singapore **January 2025 - present**

Ph.D. candidate, Medicine (GPA: 4.88/5.0)

Advisors: Prof. Jue Tao Lim, Dr. Ian Wee Liang En (Singapore General Hospital)

- Research interests: Systems Modeling, Biostatistics, Deep Learning.
- [Google Scholar].

National University of Singapore, Singapore **August 2019 - June 2023**

B.Eng. (Hons), Computer Engineering (Distinction, GPA: 4.35/5.0)

Advisors: Dr. Ganesh Neelakanta Iyer, Dr. Ravi Suppiah, Prof. Deepu John (University College Dublin)

- University Town College Program

PUBLICATIONS

1. Lin Geng*, Perran A. Ross*, Jo Yi Chow, **Zihao Wang**, Esther Li Wen Choo, Chia-Chen Chang, Xinyue Gu, Ary Hoffmann, Jue Tao Lim, "*Wolbachia*-mediated incompatible-insect technique robust to future climate change scenarios," submitted.
2. **Zihao Wang**, Dariya Nikitin, Borame L. Dickens, Liang En Wee, Martin T.W. Chio, Rayner Kay Jin Tan, Keisuke Ejima, Yi Wang, David N. Fisman, Lilith K. Whittles, and Jue Tao Lim, "Long-term public health impact of doxycycline post-exposure prophylaxis on syphilis transmission," ***Nature Health***, 2026. [Paper, GitHub].
3. **Zihao Wang**, Yuhan Li, Yao Shi, and Zhe Wu, "Input convex lipschitz recurrent neural networks for robust and efficient process modeling and optimization," arXiv. [arXiv, GitHub].
4. **Zihao Wang**, and Zhe Wu, "Towards foundation model for chemical reactor modeling: Meta-learning with physics-informed adaptation," ***Chemical Engineering Research and Design***, 218, 839-853, 2025. IF: 3.9, JCR Q2 (Engineering, Chemical). [Paper, GitHub].
5. **Zihao Wang**, Donghan Yu, and Zhe Wu, "Real-time machine-learning-based optimization using input convex long short-term memory network," ***Applied Energy***, 377, 124472, 2025. IF: 11.0, JCR Q1 (Engineering, Chemical; Energy & Fuels). [Paper, GitHub].
6. **Zihao Wang**, and Ravi Suppiah, "Upper limb movement recognition utilising EEG and EMG signals for rehabilitative robotics," *Proceedings of Future of Information and Communication Conference*, 676-695, San Francisco, California, USA, 2023. [Paper, GitHub].

   (* denotes equal contribution)

PRESENTATIONS

1. **Zihao Wang***, and Jue Tao Lim, "Modelling the interplay between doxycycline post-exposure prophylaxis, four-component meningococcal serogroup B (4CMenB) vaccine, and antimicrobial resistance in *Neisseria gonorrhoeae*," *European Congress of Clinical Microbiology and Infectious Diseases (ESCMID Global 2026)*, Munich, Germany, 2026. (Poster, **Top 2%**).
2. **Zihao Wang***, and Jue Tao Lim, "Long-run public health impact of doxycycline post-exposure prophylaxis and behavioural factors on syphilis transmission: A modelling study in Singapore and England," *Epidemics: 10th International Conference on Infectious Disease Dynamics (Epidemics10)*, San Diego, California, USA, 2025. (**Oral**).
3. **Zihao Wang***, and Jue Tao Lim, "Long-term public health impact of doxycycline post-exposure prophylaxis and behavioural factors on syphilis transmission: A modelling study in Singapore and England," *Singapore Health & Biomedical Congress (SHBC)*, Singapore, 2025. (Poster).
4. **Zihao Wang**, Wenlong Wang*, and Zhe Wu, "Accelerated modeling of various chemical processes using meta-learning-based foundation models: A few-shot learning approach using Reptile," *American Institute of Chemical Engineers (AIChE) Annual Meeting*, San Diego, California, USA, 2024. (**Oral**). [Abstract].

5. **Zihao Wang**, Wenlong Wang*, and Zhe Wu, "Input convex LSTM for fast machine learning-based optimization," *American Institute of Chemical Engineers (AIChE) Annual Meeting*, San Diego, California, USA, 2024. (Poster). [Abstract].
6. **Zihao Wang**, and Zhe Wu*, "Input convex LSTM: A convex approach for fast model predictive control," Institute for Machine Learning (Head: Professor Sepp Hochreiter), Johannes Kepler University, Linz, Austria, December 2023. (**Invited talk** by Professor Günter Klambauer).
(* denotes presenting author)

AWARDS AND HONORS

Travel Grant, ESCMID Global 2026 **04/2026**
Top Rated Abstract, ESCMID Global 2026 **04/2026**
NTU Research Scholarship, NTU **2025 - Present**
IEEE Singapore Computer Society Prize, NUS **06/2023**
Outstanding Computing Project Prize, NUS **06/2023**

DISSERTATIONS

1. **Zihao Wang**, "Multimodal sensor fusion using AI for healthcare IoT sensors," B.Eng. Dissertation, Computer Engineering, National University of Singapore, Singapore, 2023. [Paper].
2. **Zihao Wang**, "AI research for rehabilitative robotics," Undergraduate Research Opportunity Project (UROP) Thesis, Computer Engineering, National University of Singapore, Singapore, 2022. [Paper].

PROFESSIONAL EXPERIENCES

National Centre for Infectious Diseases **May 2025 - Present**
Research Staff (Part-time), Tan Tock Seng Hospital, Singapore
Advisors: Prof. Kelvin Bryan Tan, Prof. Jue Tao Lim

- Conduct research on evaluating different methodological approaches to estimate the true infection fatality ratio (IFR) during pandemics, with the aim of informing early estimation of IFR under limited data to support timely, evidence-based policy decisions, under the Programme for Research in Epidemic Preparedness and REsponse (PREPARE)
- Conduct research on population-based retrospective cohort studies for respiratory syncytial virus (RSV) and dengue

Department of Chemical & Biomolecular Engineering **July 2023 - January 2025**
Research Engineer (Full-time), National University of Singapore, Singapore
Advisor: Prof. Zhe Wu

- Conducted research on input convex neural networks, model predictive control, chemical process modeling, meta-learning, and physics-informed learning
- Wrote concept papers and full proposals for grant applications and supervised master's students

Department of Biomedical Engineering **March 2023 - July 2023**
Undergraduate Research Assistant (Internship), National University of Singapore, Singapore
Advisor: Prof. Yueming Jin

- Conducted research on computer vision for gastrectomy surgical video analysis and modality-cataract video segmentation

Rolls-Royce@NTU Corporate Laboratory **May 2022 - August 2022**
Research Engineer (Internship), Nanyang Technological University, Singapore
Advisors: Dr. Aditya Venkataraman and Dr. Benjamin Cheong

- Worked on software engineering solutions for hybrid-electric power and propulsion design tools for marine and aero engines

NUS Human-Computer Interaction (NUS-HCI) Lab **March 2022 - May 2022**
Undergraduate Research Assistant (Internship), National University of Singapore, Singapore
Advisor: Prof. Shengdong Zhao

- Conducted empirical research on human-computer interaction

SKILLS

Computer: C, C++, Java, Python, R, Assembly Language, MATLAB, PostgreSQL, Verilog, Vensim, Gephi, LaTeX
Languages: Mandarin (native), English (bilingual), Cantonese (proficient), Korean (conversational)